



NORTH SYDNEY BOYS HIGH SCHOOL

EXTENSION 2 MATHEMATICS

HSC Course Assessment Task 1

December, 2023

General instructions

- Working time – 60 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.

QUESTIONS 1-4

- Commence each new question on a NEW BOOKLET.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: # BOOKLETS USED:

Teacher (please ✓)

- ☐ Mr Berry
- ☐ Ms Lee
- ☐ Mr Umakanthan
- ☐ Dr Vranesevic

Marker's use only.

QUESTION	1	2	3	4	Total
MARKS	$\overline{12}$	$\overline{10}$	$\overline{12}$	$\overline{13}$	$\overline{47}$

Question 1 (12 Marks)		Marks
a)	If $(1 + i)z = 3 - i$, find:	
i)	z in the form $x + iy$	2
ii)	$ z $	1
iii)	$\text{Arg } z$	1
iv)	$\bar{z}$	1
b)		
i)	Find $\sqrt{8 + 6i}$	2
ii)	Solve $z^2 + (2 + 4i) - 11 - 2i = 0$	2
c)	z is a complex number such that $z = k(\cos \theta + i \sin \theta)$ where $k \in \mathbb{R}$ Show $\arg(z + k) = \frac{\theta}{2}$	3

Question 2 (10 Marks)		Marks
a)	Consider the statement “for all integers a and b , if $a + b$ is even, then a and b are even”	
i)	Write the contrapositive of the statement	1
ii)	Write the converse of the statement	1
iii)	Write the negation of the statement	1
iv)	Is the original statement true or false? Prove the statement is true/false	1
b)	State the negation of $(\forall x \in \mathbb{Z})(x = 3 \Rightarrow x^2 = 9)$	1
c)	Prove that if $n > 0$ and $4^n - 1$ is prime, then n is odd	3
d)	Prove that there are no integers x and y such that $x^2 = 4y + 2$	2

Question 3 (12 Marks)

Marks

a)

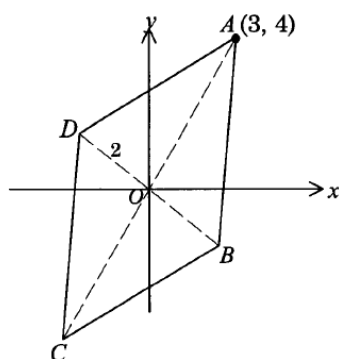


Diagram is not to scale

In the diagram, $ABCD$ is a rhombus whose diagonals meet at O , the origin. A represents the complex number $3 + 4i$ and $OD = 2$ units.

Find the complex number represented by

- | | | |
|-----|-----|---|
| i) | C | 1 |
| ii) | D | 2 |

b) The cubic equation

$$2z^3 + pz^2 + qz + 16 = 0$$

Where p and q are real, has roots α, β and γ .

It is given that $\alpha = 2 + 2\sqrt{3}i$

- | | | |
|------|--|---|
| i) | Write down another root, β , of the equation | 1 |
| ii) | Find the third root, γ | 2 |
| iii) | Find the values of p and q | 2 |
| iv) | By expressing α in modulus-argument form, show that | 2 |

$$(2 + 2\sqrt{3}i)^n = 4^n \left(\cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3} \right)$$

- | | | |
|----|------------------|---|
| v) | Hence, show that | 2 |
|----|------------------|---|

$$\alpha^n + \beta^n + \gamma^n = 2^{2n+1} \cos \frac{n\pi}{3} + \left(-\frac{1}{2} \right)^n$$

where n is an integer

Question 4 (13 Marks)

Marks

- | | | |
|-----|---|---|
| a) | Find the five roots of unity, and then express the complex roots in terms of ω , where ω is the complex root with the smallest positive argument. | 2 |
| b) | Prove that $(1 - \omega)(1 - \omega^2)(1 - \omega^3)(1 - \omega^4) = 5$ | 2 |
| c) | Show that $(1 - \omega)(1 - \omega^4) = 2 - 2 \cos \frac{2\pi}{5}$ | 2 |
| d) | Hence or otherwise, show that $\sin \frac{\pi}{5} \sin \frac{2\pi}{5} = \frac{\sqrt{5}}{4}$ | 3 |
| e) | Suppose P_0, P_1, P_2, P_3, P_4 are the corresponding points of $1, \omega, \omega^2, \omega^3$ and ω^4 in the Argand plane | |
| i) | Show that $ \overrightarrow{P_0P_1} = 2 \sin \frac{\pi}{5}$ | 2 |
| ii) | Hence find the value of | 2 |

$$|\overrightarrow{P_0P_1}| \times |\overrightarrow{P_0P_2}| \times |\overrightarrow{P_0P_3}| \times |\overrightarrow{P_0P_4}|$$

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Q1

a) If $(1+i)z = 3-i$ find z , alternately;

i) z in the form $x+iy$

$$z = \frac{3-i}{1+i} \times \frac{1-i}{1-i} \quad \checkmark \quad (1+i)z = 3-i$$

$$= \frac{3-i-3i-1}{1+1}$$

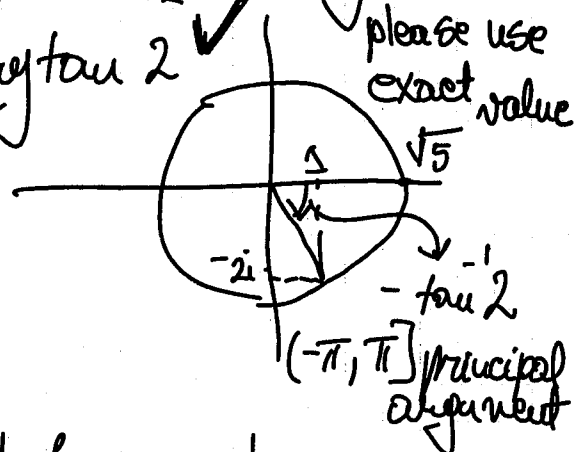
$$= \frac{2-4i}{2}$$

$$= 1-2i \quad \checkmark$$

ii) $|z| = \sqrt{1^2 + 2^2}$
 $= \sqrt{5} \quad \checkmark$

iii) $\text{Arg } z = \arg \tan \frac{-2}{1} = \arg \tan(-2)$
 $= -\arg \tan 2 \quad \checkmark$

iv) $\bar{z} = \overline{1-2i}$
 $= 1+2i \quad \checkmark$



CFE - carried forward error considered

b) Find $\sqrt{8+6i}$

$$\sqrt{8+6i} = x+iy$$

$$8+6i = x^2-y^2 + 2xyi$$

① $x^2-y^2=8$

② $2xy=6 \Rightarrow x = \frac{3}{y}$

② $\Rightarrow$ ① $\Rightarrow \frac{9}{y^2} - y^2 = 8$

$$9 - y^4 - 8y^2 = 0$$

$$y^4 + 8y^2 - 9 = 0$$

$$(y^2-1)(y^2+9) = 0$$

$$y = \pm 1, y = \pm 3i \notin \mathbb{R}$$

$$\therefore y = \pm 1 \Rightarrow x = \pm 3$$

$$\therefore \pm(3+i) \quad \checkmark$$

Alternately:

① $x^2-y^2=8$

② $x^2+y^2 = \sqrt{8^2+6^2} = 10$

①+②:
 $2x^2 = 18$
 $x = \pm 3$

②-①:
 $2y^2 = 2$
 $y = \pm 1$

$$\therefore \sqrt{8+6i} = \pm(3+i)$$

ii) Solve $z^2 + (2+4i)z - 11 - 2i = 0$

$$\Delta = b^2 - 4ac$$

$$= (2+4i)^2 + 4 \times 1 \times (11+2i)$$

$$= 4 + 16i - 16 + 44 + 8i$$

$$= 32 + 24i$$

$$= 4(8+6i)$$

$$z = \frac{-b \pm \sqrt{\Delta}}{2a}$$

$$= \frac{-(2+4i) \pm \sqrt{4(8+6i)}}{2 \times 1}$$

$$= \frac{-2-4i \pm 2(3+i)}{2} \quad \text{from (i)}$$

$$= -1-2i \pm (3+i)$$

$$= 2-i \quad \checkmark \quad \text{or} \quad -4-3i \quad \checkmark$$

c) z is complex number such as
 $z = k(\cos \theta + i \sin \theta)$, $k \in \mathbb{R}$. Show $\arg(z+k) = \frac{\theta}{2}$

Solution 1:

$$z = k(\cos \theta + i \sin \theta), k \in \mathbb{R}$$

$$z+k = k(\cos \theta + i \sin \theta) + k$$

$$= k(\cos \theta + 1) + i k \sin \theta$$

$$\arg(z+k) = \tan^{-1} \left(\frac{k \sin \theta}{k(\cos \theta + 1)} \right)$$

$$= \tan^{-1} \left(\frac{\sin \theta}{1 + \cos \theta} \right)$$

$$= \tan^{-1} \left(\frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2} - 1 + 1} \right)$$

$$= \tan^{-1} \left(\frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \right)$$

$$= \tan^{-1} \left(\tan \frac{\theta}{2} \right)$$

$$= \frac{\theta}{2}$$

$$\begin{aligned} \sin \theta &= 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \\ \cos \theta &= \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} \\ &= \cos^2 \frac{\theta}{2} - 1 + \cos^2 \frac{\theta}{2} \\ &= 2 \cos^2 \frac{\theta}{2} - 1 \end{aligned}$$

Solution 2,

$$\arg(z+k) = \arg(k(\cos\theta + i\sin\theta + 1))$$

$$= \arg(\cos\theta + i\sin\theta + 1)$$

$$= \arg(e^{i\theta} + 1)$$

$$= \arg(e^{i\theta/2} (e^{i\theta/2} + e^{-i\theta/2}))$$

$$= \arg(e^{i\theta/2} \times 2 \cos \frac{\theta}{2})$$

real part

$$= \arg e^{i\theta/2} + \arg(2 \cos \frac{\theta}{2})$$

$$= \frac{\theta}{2}$$

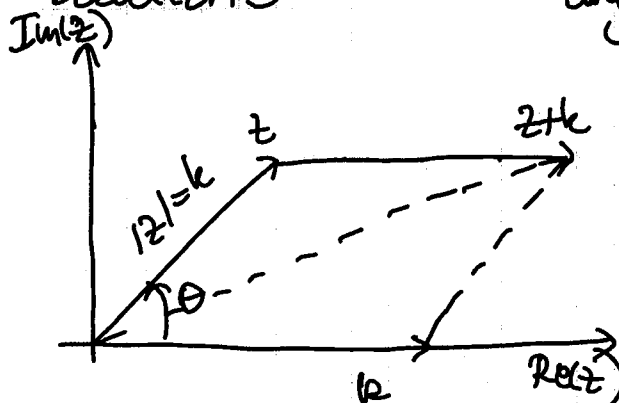
as for $z = e^{i\theta} \Rightarrow z^n = e^{in\theta} = \cos(n\theta) + i\sin(n\theta)$
 $\bar{z} = e^{-i\theta} \Rightarrow \bar{z}^n = e^{-in\theta} = \cos(n\theta) - i\sin(n\theta)$

$$e^{in\theta} + e^{-in\theta} = 2 \cos(n\theta)$$

in our case $n = \frac{1}{2}$

$$\therefore e^{i\theta/2} + e^{-i\theta/2} = 2 \cos \frac{\theta}{2}$$

Solution 3



$$\arg z = \theta$$

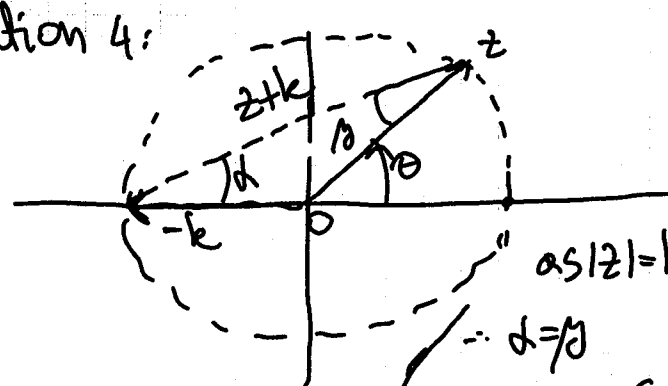
$$|z| = k$$

$$k \in \mathbb{R}$$

$z+k$ form a rhombus

$\therefore$ angle in a rhombus is bisected by its diagonals $\therefore \arg(z+k) = \frac{\theta}{2}$

Solution 4:



$$|z| = k$$

$$\arg(z) = \theta$$

$$\text{as } |z| = k \text{ \& } |-k| = k$$

$$\therefore \triangle OZ(-k) \text{ is isosceles}$$

$\angle Z = \theta$ Exterior angle theorem

$$\therefore 2h = \theta$$

$$\therefore h = \frac{\theta}{2}, \arg(z+k) = \frac{\theta}{2}$$

or: from circle geometry angle at the center is double the angle

at circumference with the same arc

Question 2

a)

- i) For all integers a and b , if a or b is odd, then $a+b$ is odd
- ii) For all integers a and b , if a and b are even, then $a+b$ is even
- iii) There exists integers a and b such that $a+b$ is even and a or b is odd
- iv) False. Counterexample: $a=3$ and $b=5$
 $a+b=8$ but neither a nor b are even.

b) $(\exists x \in \mathbb{Z})(x=3 \wedge x^2 \neq 9)$

c) The contrapositive is proven instead. That is:

If n is even, then $4^n - 1$ is not prime.

Let $n = 2k$, $k \in \mathbb{Z}$, then n is even and

$$\begin{aligned} 4^n - 1 &= 4^{2k} - 1 \\ &= (4^k + 1)(4^k - 1) \end{aligned}$$

which is a product of two integers of which neither is 1 ($n > 0$) and is thus not prime.

Hence, by the contrapositive, if $4^n - 1$ is prime, then n is odd.

d) By way of contradiction, assume there are integers x and y

such that $x^2 = 4y + 2$ (*)

Now $4y + 2 = 2(y+1)$ which is even, so x^2 is even.

Let $x = 2k$, $k \in \mathbb{Z}$

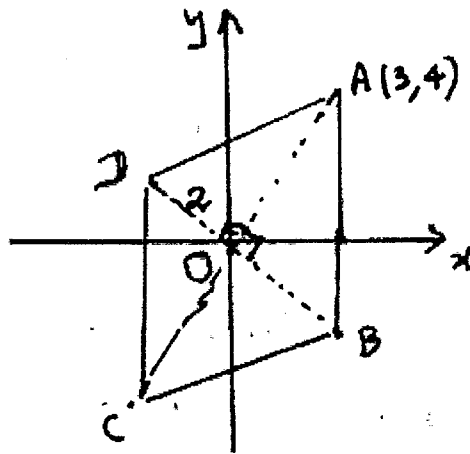
$$\begin{aligned} (2k)^2 &= 4y + 2 \\ 4k^2 &= 4y + 2 \\ 2k^2 &= 2y + 1 \end{aligned}$$

LHS is even and RHS is odd.
Even $\neq$ Odd

Thus we have a contradiction, so there must not be any integers x and y such that $x^2 = 4y + 2$

Q3

(a)



(i) Diagonals bisect in a Rhombus

$$\begin{array}{l|l} \therefore \vec{OC} = -\vec{OA} & \vec{OC} = \vec{OA} \operatorname{cis} \pi \\ = -(3+4i) & = -1(3+4i) \\ = \underline{\underline{-3-4i}} & = \underline{\underline{-3-4i}} \end{array}$$

Diagonals bisect + correct Answer for
No reason no mark 1 mark

$$(ii) |\vec{OA}| = 2, |\vec{OA}| = \sqrt{3^2+4^2} = 5$$

$$\begin{aligned} \Rightarrow \vec{OP} &= \frac{2}{5} \vec{OA} \operatorname{cis} \frac{\pi}{2} \quad \checkmark \quad (\text{diagonals } \perp \text{ or in a rhombus}) \\ &= \frac{2}{5} (3+4i)i \\ &= \underline{\underline{-\frac{8}{5} + \frac{6i}{5}}} \quad \checkmark \end{aligned}$$

(b) $2z^3 + pz^2 + qz + 16 = 0 \quad p, q \in \mathbb{R}$
 roots α, β and γ

(i) $\alpha = 2 + 2\sqrt{3}i$

Since the coefficients are all real,
 the complex conjugate of α is also a root.

let $\beta = \bar{\alpha} = \underline{2 - 2\sqrt{3}i}$ ✓ (correct answer)
 (Reason: given the benefit of doubt)

(ii) Product of roots $\alpha\beta\gamma = -\frac{16}{2} = -8$

$\Rightarrow \alpha\bar{\alpha}\gamma = -8$ ✓

$\Rightarrow [2^2 + (2\sqrt{3})^2]\gamma = -8$

$\Rightarrow \gamma = -\frac{8}{16} = -\frac{1}{2}$ ✓

(iii) Sum of roots $\alpha + \beta + \gamma = -\frac{p}{2}$

$\alpha + \bar{\alpha} + \gamma = -\frac{p}{2}$

$2(2) + (-\frac{1}{2}) = -\frac{p}{2}$

$\Rightarrow \underline{p = -7}$ ✓

~~sum~~ sum of product of 2 roots at a time = $\frac{q}{2}$

$\Rightarrow \alpha\bar{\alpha} + \bar{\alpha}\gamma + \gamma\alpha = \frac{q}{2}$

$16 + (4)(-\frac{1}{2}) = \frac{q}{2}$

$\Rightarrow \underline{q = 28}$ ✓

← (or by factorising)

(iv)

$$\alpha = 2 + 2\sqrt{3}i$$

$$= 4\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$$

$$= 4 \operatorname{cis}\left(\frac{\pi}{3}\right) \checkmark$$

$$\alpha^n = 4^n \operatorname{cis}\left(\frac{n\pi}{3}\right) \checkmark \text{ (de Moivre's theorem.)}$$

$$\therefore \beta^n = (\bar{\alpha})^n = 4^n \operatorname{cis}\left(-\frac{n\pi}{3}\right)$$

$$\begin{aligned} \text{(v)} \Rightarrow \alpha^n + \beta^n + \gamma^n &= 4^n \operatorname{cis}\left(\frac{n\pi}{3}\right) + 4^n \operatorname{cis}\left(-\frac{n\pi}{3}\right) + \left(-\frac{1}{2}\right)^n \\ &= 4^n \left[2 \cos\left(\frac{n\pi}{3}\right)\right] + \left(-\frac{1}{2}\right)^n \\ &= \underline{\underline{2^{n+1} \cos\left(\frac{n\pi}{3}\right) + \left(-\frac{1}{2}\right)^n}} \checkmark \end{aligned}$$

Question 4

a) $z^5 = 1$

$$z_1 = 1$$

$$z_2 = \cos \frac{2\pi}{5} = \omega$$

$$z_3 = \cos \frac{4\pi}{5} = \omega^2$$

$$z_4 = \cos \frac{6\pi}{5} = \cos \left(-\frac{4\pi}{5}\right) = \omega^3$$

$$z_5 = \cos \frac{8\pi}{5} = \cos \left(-\frac{2\pi}{5}\right) = \omega^4$$

b) M1 $z^5 - 1 = (z-1)(z-\omega)(z-\omega^2)(z-\omega^3)(z-\omega^4)$ ①

$$= (z-1)(z^4 + z^3 + z^2 + z + 1)$$
 ②

Equating ① & ②,

$$(z-\omega)(z-\omega^2)(z-\omega^3)(z-\omega^4) = z^4 + z^3 + z^2 + z + 1$$

Let $z = 1$:

$$(1-\omega)(1-\omega^2)(1-\omega^3)(1-\omega^4) = 1 + 1 + 1 + 1 + 1 = 5$$

M2

$$\begin{aligned} & (1-\omega)(1-\omega^2)(1-\omega^3)(1-\omega^4) \\ &= (1-\omega-\omega^2+\omega^3)(1-\omega^3-\omega^4+\omega^7) \\ &= 1 - \omega^3 - \omega^4 + \omega^7 - \omega + \omega^4 + \omega^5 - \omega^8 - \omega^2 + \omega^5 + \omega^6 \\ & \quad - \omega^9 + \omega^3 - \omega^6 - \omega^7 + \omega^{10} \\ &= 1 - \omega - \omega^2 + 2 - \omega^3 - \omega^4 + 1 \quad \text{since } \omega^5 = 1 \\ &= 4 - (\omega - \omega^2 - \omega^3 - \omega^4) \\ &= 4 - (-1) \quad \text{since } \omega^5 = 1 \text{ as } \omega \text{ is a root of } z^5 \\ &= 5 \quad \omega^5 - 1 = 0 \end{aligned}$$

$$(\omega-1)(1+\omega+\omega^2+\omega^3+\omega^4) = 0$$

$$1+\omega+\omega^2+\omega^3+\omega^4 = 0 \quad \text{since } \omega \neq 1$$

$$\omega + \omega^2 + \omega^3 + \omega^4 = -1$$

$$\begin{aligned}
 c) \quad (1-\omega)(1-\omega^4) &= (1-\cos \frac{2\pi}{5})(1-\cos(-\frac{2\pi}{5})) \quad \text{from (a)} \\
 &= 1 - (\cos \frac{2\pi}{5} + \cos(-\frac{2\pi}{5})) + \cos(\frac{2\pi}{5} + (-\frac{2\pi}{5})) \\
 &= 1 - 2\cos \frac{2\pi}{5} + \cos 0 \quad \text{since } \cos(-x) = \cos x \\
 &\quad \text{and } \sin(-x) = -\sin x \\
 &= 1 - 2\cos \frac{2\pi}{5} + 1 \\
 &= 2 - 2\cos \frac{2\pi}{5}
 \end{aligned}$$

$$\begin{aligned}
 d) \quad \text{Similarly, } (1-\omega^2)(1-\omega^3) &= 2 - 2\cos \frac{4\pi}{5} \\
 (1-\omega)(1-\omega^4)(1-\omega^2)(1-\omega^3) &= 5 \quad \text{from (6)} \\
 \therefore (2 - 2\cos \frac{2\pi}{5})(2 - 2\cos \frac{4\pi}{5}) &= 5 \\
 (2 - 2(1 - 2\sin^2 \frac{\pi}{5}))(2 - 2(1 - 2\sin^2 \frac{2\pi}{5})) &= 5 \quad \text{since } \cos 2x = 1 - 2\sin^2 x \\
 4\sin^2 \frac{\pi}{5} \times 4\sin^2 \frac{2\pi}{5} &= 5 \\
 (\sin \frac{\pi}{5} \sin \frac{2\pi}{5})^2 &= \frac{5}{16}
 \end{aligned}$$

$$\therefore \sin \frac{\pi}{5} \sin \frac{2\pi}{5} = \sqrt{\frac{5}{16}} = \frac{\sqrt{5}}{4}$$

$\sin \frac{\pi}{5}, \sin \frac{2\pi}{5} > 0$

$$\begin{aligned}
 e) i) M1 \quad |\vec{P_0 P_1}| &= |\omega - 1| \\
 &= |\cos \frac{2\pi}{5} - \cos 0| \\
 &= |\cos \frac{\pi}{5} (\cos \frac{\pi}{5} - \cos(-\frac{\pi}{5}))| \\
 &= |\cos \frac{\pi}{5} (2\sin \frac{\pi}{5})| \\
 &= |\cos \frac{\pi}{5}| |2\sin \frac{\pi}{5}| \\
 &= 1 \times 2 \sin \frac{\pi}{5}
 \end{aligned}$$

M2 Using the cosine rule.

$$\begin{aligned}
 |\vec{P_0 P_1}| &= \sqrt{1^2 + 1^2 - 2(1)(1)\cos \frac{2\pi}{5}} \\
 &= \sqrt{2 - 2\cos \frac{2\pi}{5}} \\
 &= \sqrt{2 - 2(1 - 2\sin^2 \frac{\pi}{5})} \quad \text{since } \cos 2x = 1 - 2\sin^2 x \\
 &= \sqrt{4\sin^2 \frac{\pi}{5}} \\
 &= 2\sin \frac{\pi}{5}
 \end{aligned}$$

ii) Similarly, $|\vec{P_0 P_2}| = 2 \sin \left(\frac{4\pi}{5} \times \frac{1}{2} \right) = 2 \sin \frac{2\pi}{5}$

$$|\vec{P_0 P_3}| = \left| 2 \sin \left(-\frac{4\pi}{5} \times \frac{1}{2} \right) \right| = 2 \sin \frac{2\pi}{5}$$

$$|\vec{P_0 P_4}| = \left| 2 \sin \left(-\frac{2\pi}{5} \times \frac{1}{2} \right) \right| = 2 \sin \frac{\pi}{5}$$

$$|\vec{P_0 P_1}| \times |\vec{P_0 P_2}| \times |\vec{P_0 P_3}| \times |\vec{P_0 P_4}|$$

$$= \left(2 \sin \frac{2\pi}{5} \times 2 \sin \frac{\pi}{5} \right)^2$$

$$= 16 \times \left(\frac{\sqrt{5}}{4} \right)^2$$

$$= 5$$